MXET 375
Applied Dynamic Systems



Multidisciplinary
Engineering Technology
COLLEGE OF ENGINEERING

Final Project

3 DOF Spring Loaded Robotic Finger

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Introduction

The purpose of this project is to accurately design, mathematically model, simulate, and gather experimental data for a three degree-of-freedom spring loaded robotic finger that is driven by high-torque servo motors. This project was designed with the intention of imitating the bending motion of a physical human finger by offering independent control of each joint before returning to its original position. This finger design utilizes a pair of springs behind each joint to provide a reflexive restoring torque which would allow the system to return back to its upright position without additional motor actuation. Ultimately this project would be chosen for displaying complex understanding of mathematically modeling dynamic systems, simulating a complex system, and using conceptual understanding coupled with empirical evaluation to validate findings of a multi-bodied dynamic system. These desired goals would be accomplished through using Onshape CAD software to design the physical prototype, Arduino software to control the servo output and collect current data, Simulink to simulate the expected behavior of the system, and a combination of python libraries to track the position of brightly colored stickers on each joint of the finger which included NumPy and Pandas. The final expected outcomes of this project would include a fully functional physical prototype, in which independent joint actuation is performed along with gathering joint position data from recorded trials. Important theoretical concepts to mention would include the proportional relationship between torque and current. Since servos don't have the ability to directly produce and output torque or directly offer the ability to measure the torque applied during its motion, it's important to create an estimation of the torque output of the servos as an input into the system. This estimation would be accomplished by assuming a proportional relationship between torque and output current of the servos. Using this understanding of the relationship between physical variables to create an accurate estimate of the inputs into the system, an accurate model can be developed that can be further supported with the quantifiable results of this project. Lastly, verify the potential reason for the discrepancies that may arise between the data collected from the simulated joint angle and the experimental joint angle.

Background and Dynamic Modeling

The robotic finger consists of three rigid phalange segments that are connected in series by rotational joints. Each joint represents one rotational degree of freedom, which results in a 3 degree-of-freedom serial multi-linked chain. A high-torque servo drives each joint by applying a unique input rotational distance to a tendon which is attached to said joint. At the same time the two opposing springs create a restoring torque to assist in returning the finger to its neutral position when the servo input acts in the opposite direction. Each joint experiences motor torque, spring restoring torque and resistive loads such as friction and inertia. The free body diagrams in Figures 1 and 2 show the forces acting on each segment, and the torques applied to each joint in the system respectively.

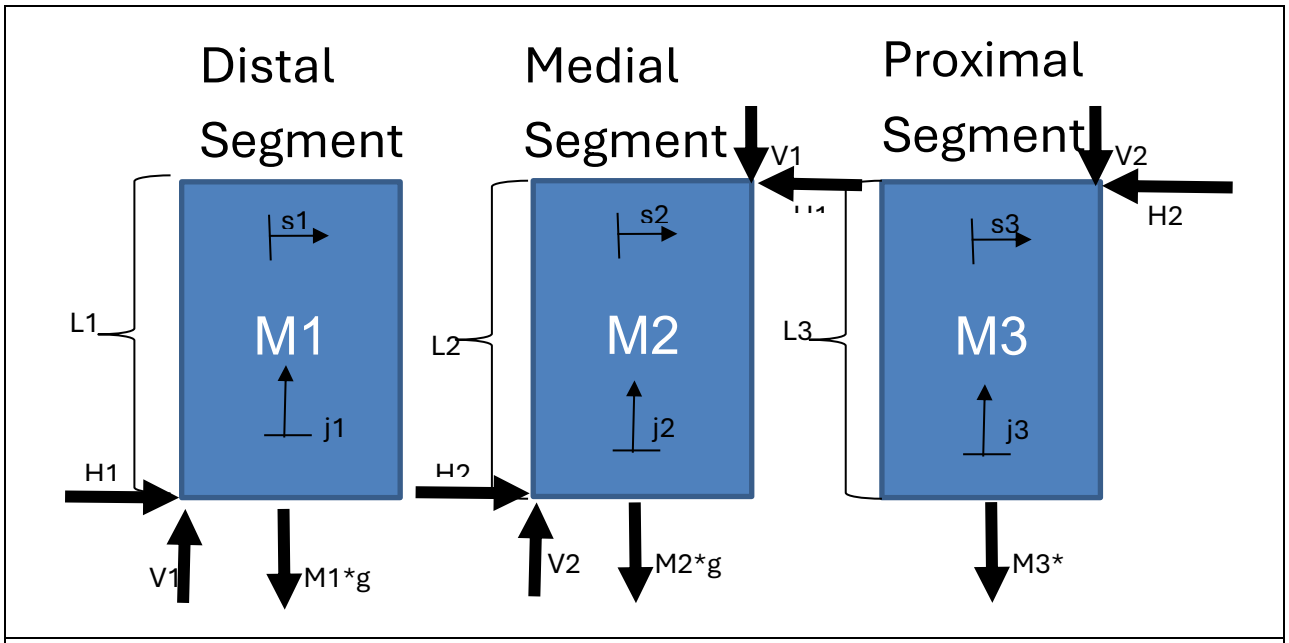


Figure 1: Free Body Diagram of Each Segment of the Finger

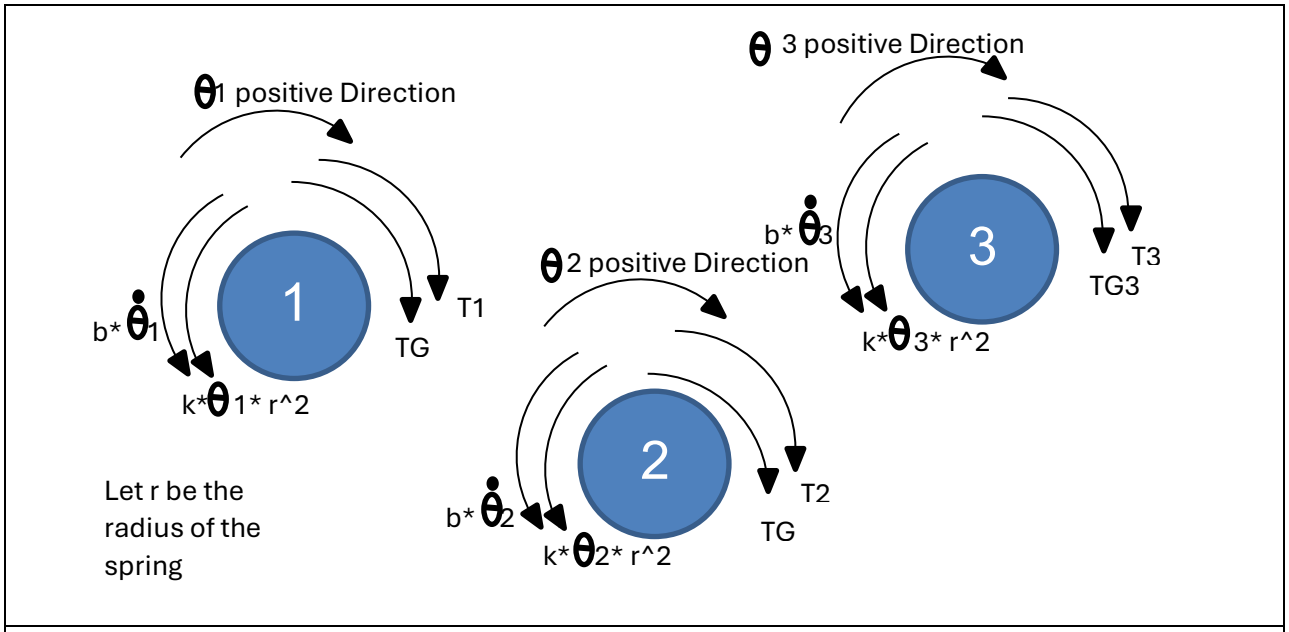


Figure 2: Free Body Diagram of Each Joint of the Finger

The dynamic behavior of each joint can be explained using rotational equations of motion created based on the techniques learned in this course. These equations will explain the relation of torque due to gravity, applied servo torque, spring forces, damping, and friction.

The following will include the derivation of the equations of motion for segment 1, the distal segment.

$$J_1\theta_1'' = T_1 + T_{G1} - k\theta_1 r^2 - b\theta_1' + V_1\sin(\theta_1) + H_1\cos(\theta_1)$$

$$m_1 x_1'' = H_1$$

$$m_1 x_{G1}'' = H_1$$
$$m_1 y_{G1}'' = V_1 - m_1 g$$

$$x_{G1} = s_1 + L_1 \sin(\theta_1)$$
$$x_{G1}' = s_1' L_1 \theta_1' \cos(\theta_1)$$
$$x_{G1}'' = s_1'' + L_1 \theta_1'' \cos(\theta_1) - L_1 (\theta_1')^2 \sin(\theta_1)$$

$$y_{G1} = j_1 + L_1 \cos(\theta_1)$$
$$y_{G1}' = j_1' - L_1 \theta_1' \sin(\theta_1)$$
$$y_{G1}'' = j_1'' - L_1 (\theta_1')^2 \cos(\theta_1) - L_1 \theta_1'' \sin(\theta_1)$$
$$H_1 = m_1 (s_1'' + L_1 \theta_1'' \cos(\theta_1) - L_1 (\theta_1')^2 \sin(\theta_1))$$
$$V_1 = m_1 (j_1'' - L_1 (\theta_1')^2 \cos(\theta_1) - L_1 \theta_1'' \sin(\theta_1) + g)$$

Equation 1

$$J_1 \theta_1'' = T_1 + T_{G1} - k\theta_1 r^2 - b\theta_1' + m_1 (j_1'' - L_1 (\theta_1')^2 \cos(\theta_1) - L_1 \theta_1'' \sin(\theta_1) + g) \sin(\theta_1)$$
$$+ m_1 (s_1'' + L_1 \theta_1'' \cos(\theta_1) - L_1 (\theta_1')^2 \sin(\theta_1)) \cos(\theta_1)$$

Equation 2

$$m_1 x_1'' = m_1 (s_1'' + L_1 \theta_1'' \cos(\theta_1) - L_1 (\theta_1')^2 \sin(\theta_1))$$

The following will include the derivation of the equations of motion for segment 2, the medial segment.

$$J_2 \theta_2'' = T_2 + T_{G2} - k\theta_2 r^2 - b\theta_2' + V_2 \sin(\theta_2) + H_2 \cos(\theta_2) - V_1 \sin(\theta_1) - H_1 \cos(\theta_1)$$

$$m_2 x_2'' = H_2 - H_1 = m_2 x_{G2}''$$
$$m_2 y_2'' = V_2 - V_1 - m_2 g$$

$$x_{G2} = s_2 + L_2 \sin(\theta_2)$$
$$x_{G2}' = s_2' + L_2 \theta_2' \cos(\theta_2)$$
$$x_{G2}'' = s_2'' + L_2 \theta_2'' \cos(\theta_2) - L_2 (\theta_2')^2 \sin(\theta_2)$$

$$y_{G2} = j_2 + L_2 \cos(\theta_2)$$
$$y_{G2}' = j_2' - L_2 \theta_2' \sin(\theta_2)$$
$$y_{G2}'' = j_2'' - L_2 \theta_2'' \sin(\theta_2) - L_2 (\theta_2')^2 \cos(\theta_2)$$

$$H_2 = m_2 x_{G2}'' + H_1$$
$$V_2 = m_2 y_{G2}'' + V_1 + m_2 g$$

Equation 3

$$J_2 \theta_2'' = T_2 + T_{G2} - k\theta_2 r^2 - b\theta_2'$$

$$+\big(m_2(j_2'' - L_2\theta_2'' \sin(\theta_2) - L_2(\theta_2')^2 \cos(\theta_2) + g)$$
$$+ m_1(j_1'' - L_1(\theta_1')^2 \cos(\theta_1) - L_1\theta_1'' \sin(\theta_1) + g)\big)\sin(\theta_2)$$
$$+ \big(m_2(s_2'' + L_2\theta_2'' \cos(\theta_2) - L_2(\theta_2')^2 \sin(\theta_2))$$
$$+ m_1(s_1'' - L_1(\theta_1')^2 \sin(\theta_1) + L_1\theta_1'' \cos(\theta_1))\big)\cos(\theta_2)$$
$$- \big(m_1(j_1'' - L_1(\theta_1')^2 \cos(\theta) - L_1\theta_1'' \sin(\theta_1) + g)\big)\sin(\theta_1)$$
$$- \big(m_1(s_1'' - L_1(\theta_1')^2 \sin(\theta) + L_1\theta_1'' \cos(\theta_1))\big)\cos(\theta_1)$$

Equation 4

$$m_2 x_2'' = m_2(s_2'' + L_2\theta_2'' \cos(\theta_2) - L_2(\theta_2')^2 \sin(\theta_2))$$

The following will include the derivation of the equations of motion for segment 3, the proximal segment.

$$J_3\theta_3'' = T_3 + T_{G3} - k\theta_3 r^2 - b\theta_3' - V_2 \sin(\theta_3) - H_2 \cos(\theta_3)$$
$$m_3 x_3'' = -H_2$$

Equation 5

$$J_3\theta_3'' = T_3 + T_{G3} - k\theta_3 r^2 - b\theta_3'$$
$$-\big(m_2(j_2'' - L_2\theta_2'' \sin(\theta_2) - L_2(\theta_2')^2 \cos(\theta_2) + g)$$
$$+ m_1(j_1'' - L_1\theta_1'' \sin(\theta_1) - L_1(\theta_1')^2 \cos(\theta_1))\big)\sin(\theta_3)$$
$$-\big(m_2(s_2'' + L_2\theta_2'' \sin(\theta_2) - L_2(\theta_2')^2 \sin(\theta_2)) + m_1(s_1'' + L_1\theta_1'' \cos(\theta_1) - L_1(\theta_1')^2 \sin(\theta_1))\big)\cos(\theta_3)$$

Equation 6

$$m_3 x_3'' = -\big(m_2(s_2'' + L_2\theta_2'' \sin(\theta_2) - L_2(\theta_2')^2 \sin(\theta_2)) + m_1(s_1'' + L_1\theta_1'' \cos(\theta_1) - L_1(\theta_1')^2 \sin(\theta_1))\big)$$

This is a system–level diagram that shows the full 3 degree-of-freedom object, spring placement, distance between parts, and servo inputs.

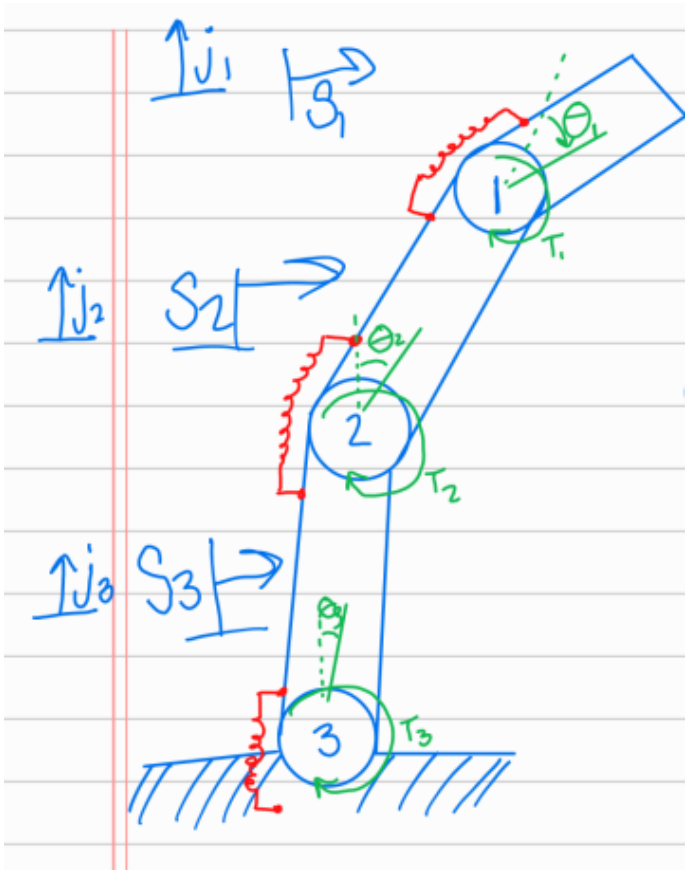


Figure 3: System–Level Diagram

Simulations and Experimental Setup

The dynamic system was simulated using MATLAB in the Simulink and Simscape Multibody software. The simulation is replicating a 3 degree-of-freedom object, servo motor actuation, spring forces joint constraints and expected positions. The simulated model was constructed to mimic the physical behavior of the prototype by representing each phalange as a rigid body connected through the revolute joints with added motor torques.

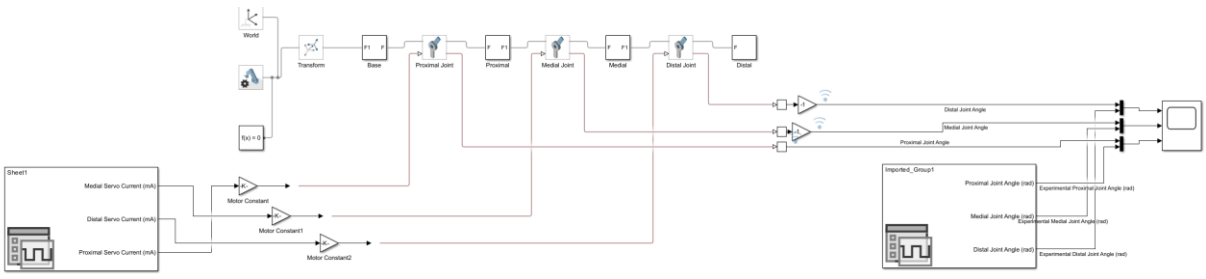


Figure 4: Block Diagram of the System

The block diagram shown in Figure 4 was used to run the simulation. The model consisted of a World block which defines the global reference frame and gravity of the whole model, the Transform block which sets a fixed position or rotation between two frames to accurately organize the model components, the Gain blocks which convert the measured servo currents into torques by multiplying by a torque constant. In addition, there are four rigid bodies representing the proximal, medial, and distal segments of the finger, as well as a rigid base. Each part is connected to a revolute joint block which allows for rotation along an axis equivalent to the physical design. Each revolute joint is labelled according to the part of the finger it is attached to where the proximal joint is attached to the proximal part, the medial joint is attached to the medial part, and the distal joint is attached to the distal part.

The motor torques were applied to each of the joints using the recorded servo currents. The servo current signals were imported from the data sheet block and then multiplied by a motor torque constant to gauge the torque acting at each joint. The torque inputs were then directly connected to the appropriate revolute joint block to imitate the experimental situation. For the comparison of theoretical to experimental data, the experimental position data was added to the signal editor.

The joint blocks output angular position signals which were first connected to the gain block and then sent to the scope for display. This was so that the simulated proximal, medial, and distal joint angles were correctly compared with the experimental tracking data. The parameter evaluation of the theoretical vs experimental position data can be seen in Figure 5.

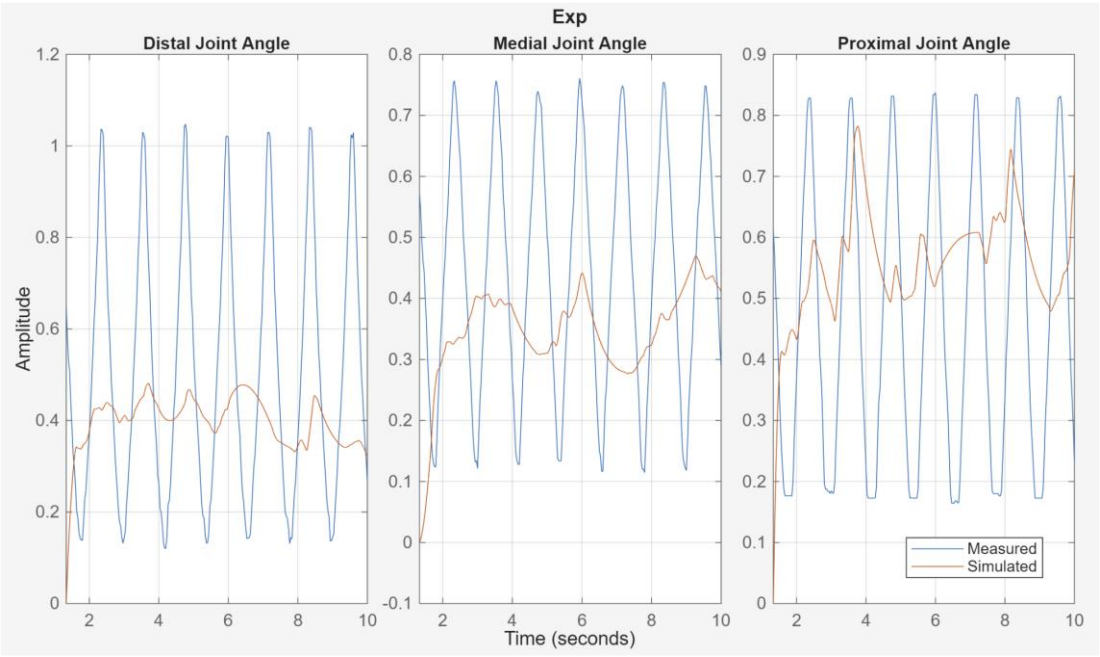


Figure 5: Parameter Evaluation of Each Joint

For empirical testing, a physical robotic finger was created using 3D-printed parts, revolute joints, three high-torque sensors, and a pair of springs at each joint. Colored tracking markers will be

attached at each joint to allow a motion-tracking software platform to detect and record the joint position in real time. Just as shown in Figure 6.



Figure 6: Fully assembled Finger with colored markers and loaded springs

The above image in Figure 6 shows the fully assembled finger with the colored marker attached to each joint. The physical system is designed to allow each joint to rotate independently by a MG 996R high torque servo motor with a pair of loaded springs attached to each joint to return the finger to its original position. An Arduino Uno microcontroller was used as the main device used to control and send commands to the servos as well as logging the position data. The Arduino generated PWM control signals for each of the three servos and recorded electrical measurements during each actuation. The system was powered using a battery pack to power the high-torque servos. Each servo was mounted at the side of the motor mount with the actuator directly underneath the modeled finger to provide easier access to the joints, creating less friction and better power access. Pairs of springs were attached to each joint, with 15mm springs for the medial and distal joints, and 10mm springs for the proximal joints. For motion tracking, a recording of the full system moving to a set desired location and returning back to its initial state. The .mp4 video was processed using the NumPy and Pandas Python libraries to extract the joint angle data from its motions. Each joint actuated individually, and the Arduino stored the time stamped measurements which includes the current and the input angles. The resulting final physical prototype is displayed in Figure 7.

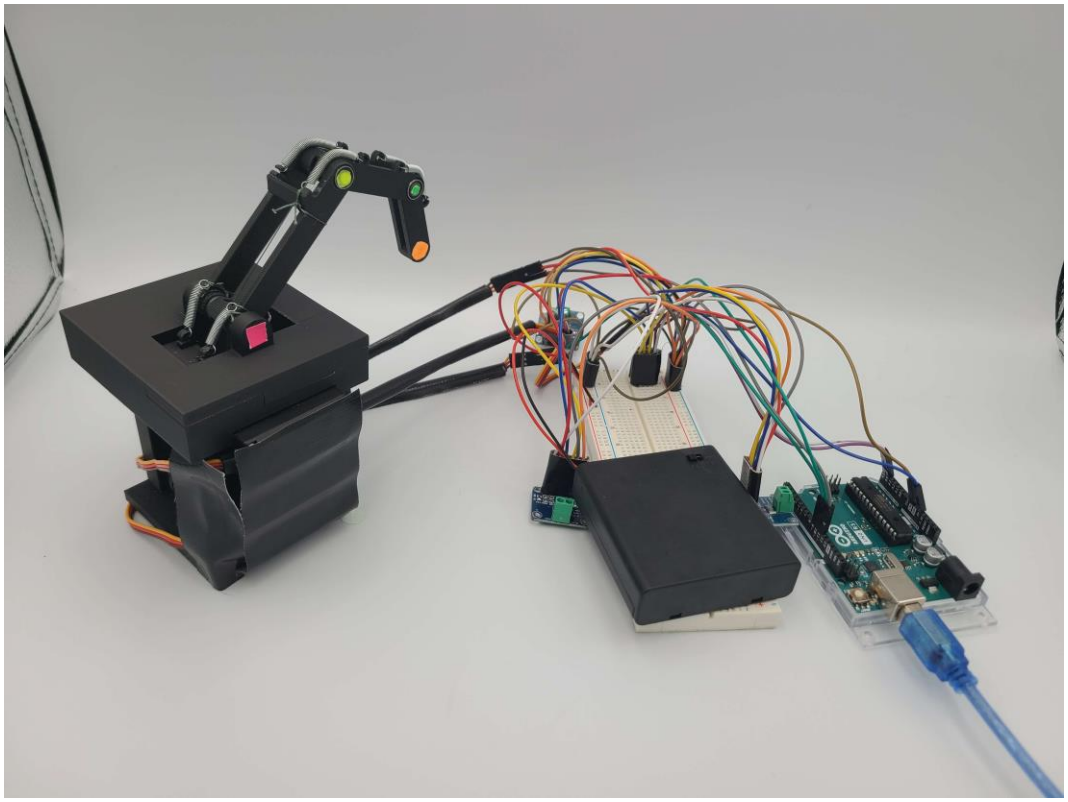


Figure 7: Final Prototype Including Circuit Connections

Evaluation of Experimental Results

The performance of the finger joint angle data was checked by comparing the simulated joint positions with the experimentally measured angles that were collected through the colored tracking system software. This was a unique data set not previously used for parameter estimation. Figure 8 shows the simulated and experimental joint angles for the distal, medial, and proximal joints over a period of five seconds.

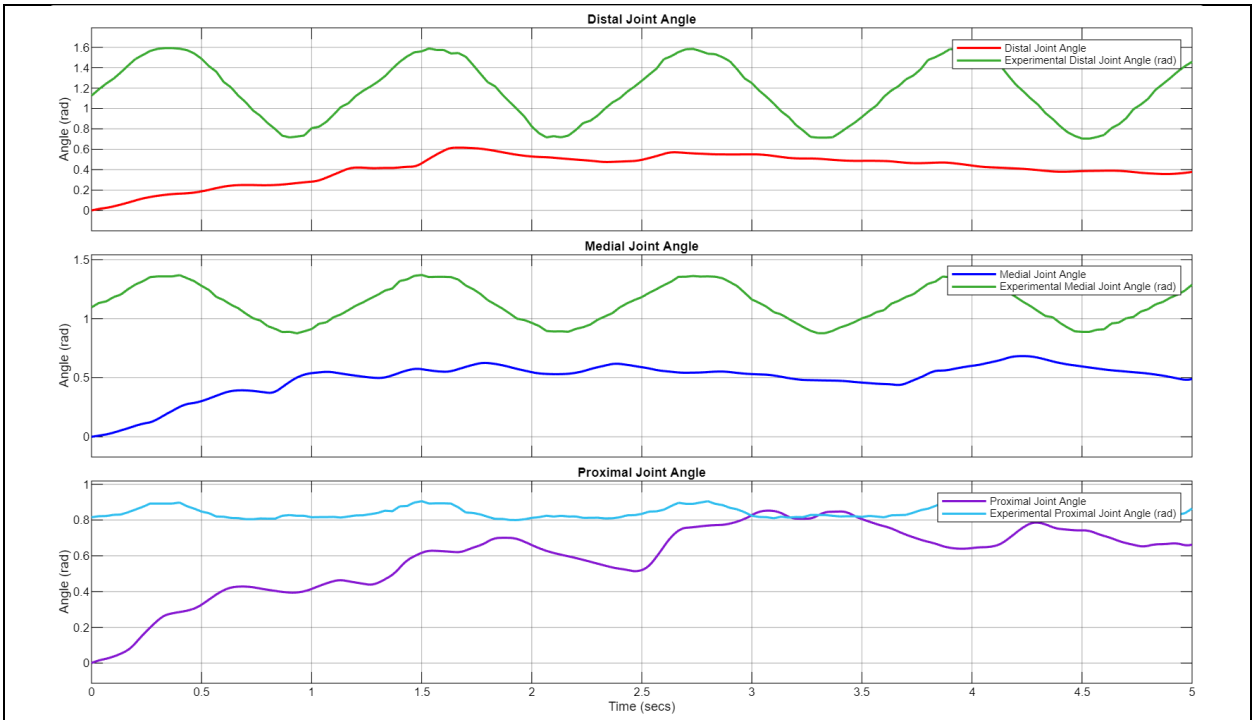


Figure 8: Predicted vs. Experimental Joint Data

Throughout all three joints, the simulated angles were much smoother and more consistent when compared to the experimental measurements. The experimental joint angles showed more oscillating behavior with much higher amplitude. The difference between the two can be explained that the experimental physical system most likely experienced unknown dynamic effects that weren't expressed in the simulated model such as an uneven spring compression or deformation of the 3D printed part when under load.

Starting from the distal joint, the experimental angle curve shows a repeated oscillation with the amplitude ranging from about 0.7 to 1.6 rad while the simulated is around 0 to 0.6 rad. For the medial joint, the experimental angle curve also shows a repeated oscillation with the amplitude ranging from 0.9 rad to 1.4 rad while the simulated joint ranges from 0.5 to 0.6 rad. For the proximal joint, the difference is a lot smaller but still present with the experimental angle curve showing a repeated oscillation with the amplitude ranging from 0.8 to 0.9 rad while the simulated joint is at 0.6 to 0.7 rad.

Overall, the experimental measurements showed higher levels of oscillations with much higher peaks when compared to the simulated joint angles. The main causes for such discrepancies could be the spring dynamics, joint friction, or 3D printed deformations.

Conclusion

In conclusion, the three degree-of-freedom spring–loaded finger demonstrated a multidisciplinary integration of mechanical design, servo actuations, dynamic modeling, and motion tracking. This project demonstrated the dynamic behavior of a multi-joint system and showed the importance of accurately modeling torques and actuator response. When comparing the simulated and experimental joint angles with one another, it reinforces the dynamic concepts taught in the course such as the rotational inertia, torque-current relationships, and forces in a spring. These concepts provided a necessary understanding to make general predictions of how the whole system would behave and operate, even though the experimental data showed a much greater amplitude of oscillations. Several challenges arose in building and refining the system, which included spring selection, reaching consistent motor behavior, reducing friction at joints, proper dimensioning, and collecting well-documented motion-tracking data. However, the largest issue with the design was the choice of servos over BLDC motors. BLDC motors draw a steady DC current with relatively little fluctuations, while servo motors have inherent noise in their current draw because of their noise internal position sensing. This resulted in the collected current data having a significant amount of noise. If BLDC motors are unavailable, this could also be partially counteracted by placing a bandpass filter in series with the current sensors. The Simulink model could alternatively be expanded to include a bandpass filter on the current data.

The concepts learned during this project can be applied in different areas such as robotics, prosthetics, system identification, and dynamic system design. Understanding the behavior of a multi-degree-of-freedom system is fundamental and important for engineering applications that involve compliant mechanisms, manipulators, and human-robot interaction.